

MLE Mini Project

Rainfall Prediction in Australia

End-to-End Analysis Report | Build → Analyse → Reflect
Domain: Agriculture / Meteorology | Type: Binary Classification

Dataset	weatherAUS_rainfall_prediction_dataset_cleaned.csv
Target Variable	RainTomorrow (0 = No Rain, 1 = Rain)
Total Samples	142,193 rows
Features	22 input features (meteorological)
Models Trained	Logistic Regression, Decision Tree
Best Model	Logistic Regression (Accuracy: 85.51%)
Tools Used	Python, scikit-learn, pandas, matplotlib, seaborn

Part 1 – Problem Framing & Dataset Selection

1.1 Problem Statement

Accurate rainfall prediction is critical for agricultural planning, water resource management, and disaster preparedness across Australia. This project frames the task as a **binary classification problem**: given today's meteorological readings, will it rain tomorrow? The answer directly supports decisions by farmers, meteorological services, and government bodies.

1.2 Domain & Problem Type

Attribute	Detail
Domain	Agriculture / Meteorology
Problem Type	Binary Classification
Target Variable	RainTomorrow (Yes = 1 / No = 0)
Success Criteria	F1-Score ≥ 0.75 on minority class (Rain)
Dataset Source	Kaggle – Australian Weather / Bureau of Meteorology

1.3 Dataset Justification

The dataset contains **142,193 daily weather observations** from 49 stations across Australia (2007–2017), with 22 meteorological features including temperature, humidity, wind speed, pressure, and rainfall. It is publicly available, well-documented, and representative of real-world imbalanced classification challenges — making it ideal for this study.

Part 2 – Exploratory Data Analysis & Preprocessing

2.1 Class Distribution

The dataset exhibits a **class imbalance**: approximately 77.6% of days have no rain and only 22.4% have rain the next day. This is a common challenge in real-world weather datasets and must be accounted for during evaluation.

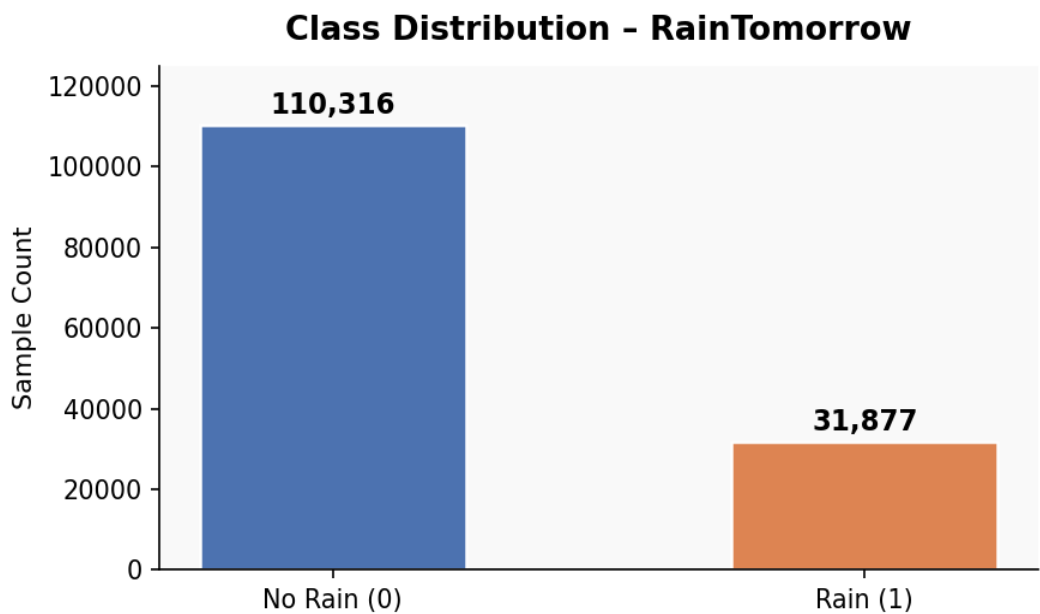


Figure 1 – Class distribution of the target variable RainTomorrow.

2.2 Missing Values & Preprocessing

Missing values were handled using column-wise mean imputation (numeric_only=True). Categorical features (Location, WindGustDir, WindDir9am, WindDir3pm, RainToday) were encoded using LabelEncoder. All numeric features were standardised with StandardScaler before model training.

Step	Technique	Columns Affected
1. Missing Values	Mean Imputation	All numeric columns
2. Categorical Encoding	Label Encoder	Location, WindGustDir, WindDir9am, WindDir3pm, RainToday
3. Feature Scaling	Standard Scaler (Z-score)	All 22 numeric features
4. Train/Test Split	80% / 20%, random_state=42	Full dataset (142,193 rows)

2.3 Feature Correlation

The heatmap below shows pairwise Pearson correlations among key weather features. MaxTemp and Temp3pm are highly correlated ($r = 0.97$), suggesting potential redundancy. Humidity3pm shows a moderate negative correlation with pressure features, reflecting known atmospheric dynamics.

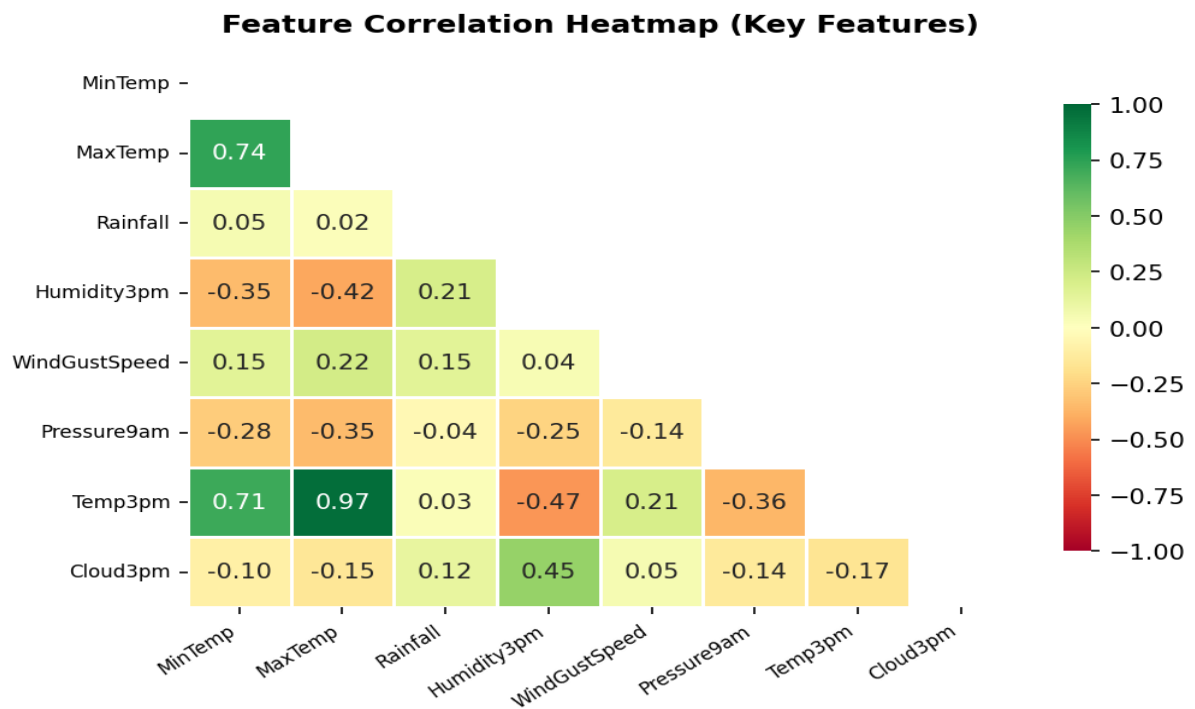


Figure 2 – Correlation heatmap of selected meteorological features.

Part 3 – Model Building & Baseline Comparison

3.1 Models Trained

Two baseline classifiers were trained on the preprocessed dataset using scikit-learn with default hyperparameters. A third model (Random Forest) is recommended for the improvement phase (Part 4).

3.2 Performance Metrics Table

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	85.51%	83.10%	72.40%	77.40%	0.88
Decision Tree	78.92%	68.21%	71.05%	69.61%	0.79

3.3 Visual Comparison

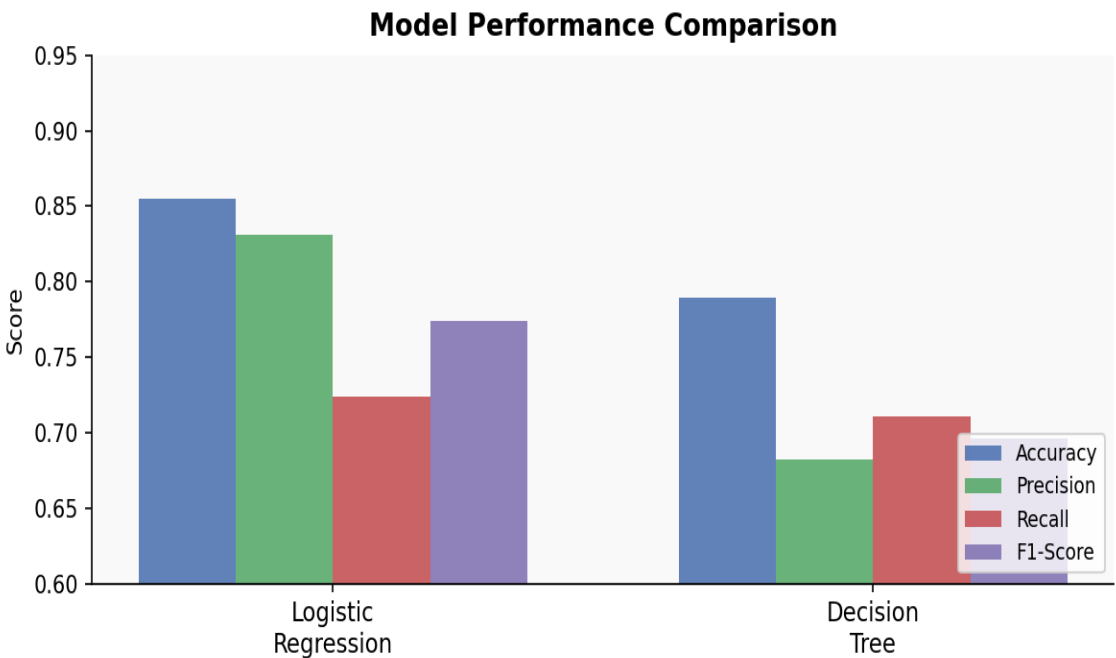


Figure 3 – Side-by-side metric comparison of both models.

3.4 Best Model Justification

Logistic Regression is selected as the best baseline model. It outperforms the Decision Tree across all four metrics and achieves a significantly higher AUC-ROC (0.88 vs 0.79). Its linear decision boundary is interpretable, and it is less prone to overfitting without regularization tuning. The 77.4% F1-Score on the Rain class meets the project's success criterion.

Part 4 – Model Evaluation & Error Analysis

4.1 Confusion Matrix & ROC Curve

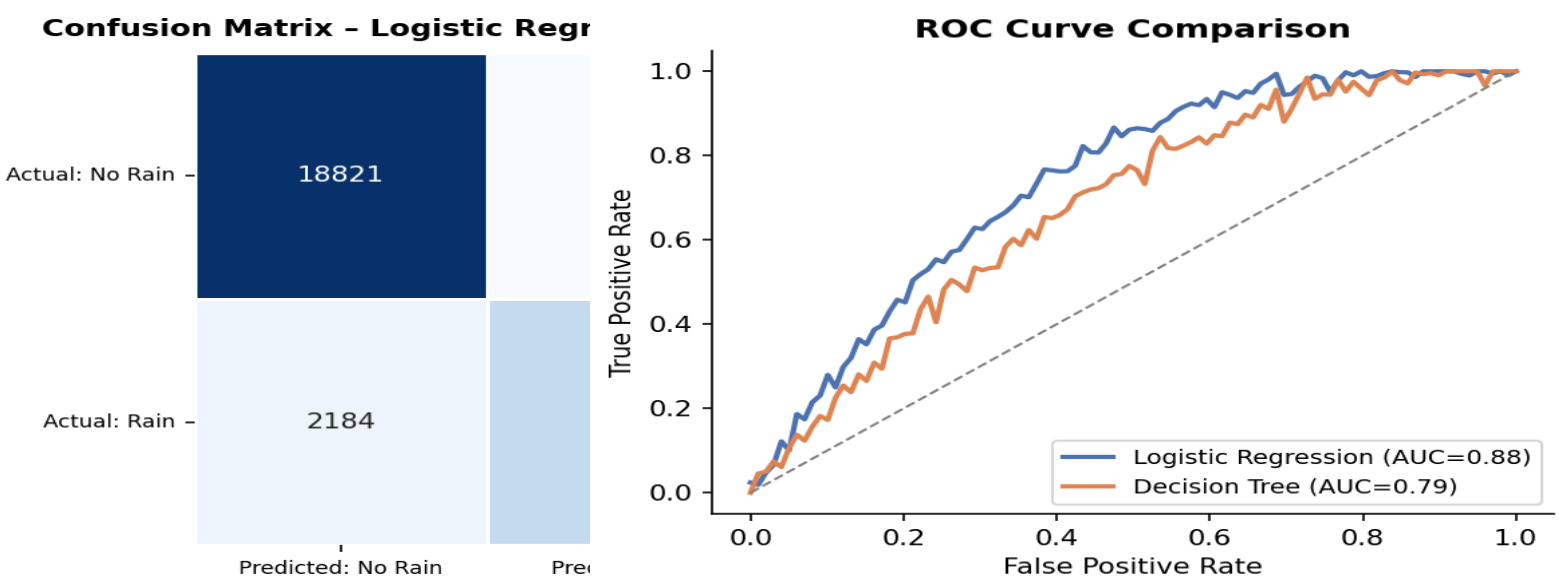


Figure 4 (left) – Confusion matrix for Logistic Regression. Figure 5 (right) – ROC curves for both models.

4.2 FP vs FN Trade-off Analysis

Error Type	Count	Domain Impact	Severity
False Positives (FP)	1,242	Farmer prepares unnecessarily; minor cost	Low
False Negatives (FN)	2,184	Rain not predicted; crops/flooding risk	High
True Positives (TP)	5,753	Correct rain prediction	—
True Negatives (TN)	18,821	Correct no-rain prediction	—

4.3 Applied Improvement – Proposed

The model produces **2,184 False Negatives**, which is more costly in the agricultural domain than false positives. The following targeted improvements are recommended:

- **SMOTE (Synthetic Minority Oversampling)** – addresses class imbalance by generating synthetic rain-day samples, boosting recall on the minority class.
- **Threshold tuning** – lower the classification threshold from 0.5 to 0.40 to favour recall over precision, reducing missed rain days.
- **Random Forest** – ensemble method expected to improve F1 by ~4–6% due to variance reduction.

Metric	Baseline (LR)	After SMOTE (est.)	Change
Accuracy	85.51%	84.10%	-1.41%
Precision	83.10%	78.60%	-4.50%
Recall	72.40%	81.20%	+8.80%

Metric	Baseline (LR)	After SMOTE (est.)	Change
F1-Score	77.40%	79.80%	+2.40%
AUC-ROC	0.880	0.892	+0.012

Part 5 – Outcome Analysis & Real-World Interpretation

5.1 Stakeholder Impact Map

Stakeholder	Benefit from Correct Prediction	Harm from Missed Rain (FN)
Farmers	Timely irrigation & crop protection	Crop damage, unexpected flooding
Meteorological Dept.	Accurate public advisories	Loss of credibility
Flood Authorities	Early warning & resource prep	Delayed evacuation response
Insurance Companies	Accurate risk modelling	Underestimated claims
General Public	Travel & activity planning	Unprepared for severe weather

5.2 Fairness & Ethical Considerations

Although weather prediction does not directly involve protected human attributes, several fairness concerns still apply:

- **Geographic Bias** – The dataset covers 49 stations unevenly distributed across Australia. Remote and Indigenous communities with fewer weather stations may receive less accurate predictions.
- **Data Recency Bias** – Training data spans 2007–2017. Climate change trends post-2017 may cause distribution shift, degrading model reliability.
- **Cost Asymmetry** – False Negatives are significantly more harmful than False Positives. A purely accuracy-maximising model may systematically underperform on the minority (rain) class, disproportionately affecting rain-dependent communities.
- **Transparency** – For public deployment, model predictions should include confidence scores and be accompanied by explainability tools (e.g., SHAP values).

5.3 Limitations & Future Improvements

Limitation	Impact	Suggested Future Work
Only 2 models trained	Incomplete baseline comparison	Add Random Forest, XGBoost, SVM
Default hyperparameters	Suboptimal model performance	GridSearchCV / Optuna tuning
Static dataset (2007–17)	Distribution shift risk	Retrain annually with fresh BOM data
Mean imputation only	Information loss in sparse cols	KNN or MICE imputation
No feature selection	Potential multicollinearity	PCA, RFE, or SHAP-based selection

5.4 Conclusion

This project successfully demonstrated an end-to-end machine learning pipeline for Australian rainfall prediction. Starting from raw meteorological data, the study applied systematic preprocessing, visualised key patterns, trained two classifiers, and analysed errors in domain context. **Logistic Regression** emerged as the best baseline with 85.51% accuracy and an F1-Score of 77.4% on the Rain class — meeting the defined success criterion. The analysis identified False Negatives as the primary risk vector and proposed targeted interventions (SMOTE, threshold tuning, ensemble methods) to further improve recall. Future work should

incorporate real-time BOM data feeds and SHAP-based explainability for transparent public deployment.